

It is a cold, windless morning and three hot-air balloons hover above a park. Each balloon is stationary and in direct line of sight, with no obstacles near them, as shown in Figure 3. Balloon A is equipped with a 100 W siren, which emits a 2000 Hz tone uniformly in all directions. On board balloons B and C are students with sound measuring equipment.

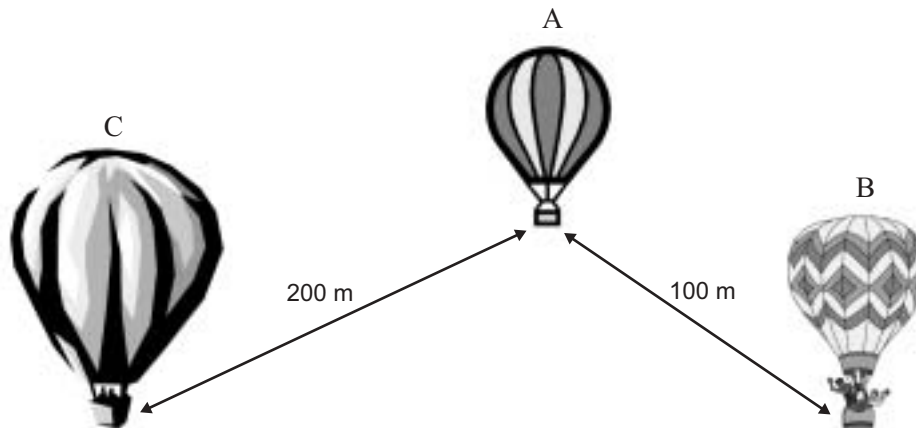


Figure 3

**Question 9**

Which of the following is the best estimate of the sound **intensity** of the siren as measured at balloon B?

- A.  $0.5 \text{ W m}^{-2}$
- B.  $2.5 \times 10^{-2} \text{ W m}^{-2}$
- C.  $8.0 \times 10^{-4} \text{ W m}^{-2}$
- D.  $2.5 \times 10^{-5} \text{ W m}^{-2}$

2 marks

**Question 10**

By how many decibels will the sound intensity level at balloon C be lower than at balloon B?

 dB

2 marks

**Question 11**

Balloons B and C move so that they are at equal distances from balloon A.

The sound intensity at balloon C is now measured as  $1.0 \times 10^{-2} \text{ W m}^{-2}$ .

What is the sound intensity level (dB) at balloon B?

 dB

2 marks

# **PHYSICS**

## **Written examination 2**

### **DATA SHEET**

#### **Directions to students**

Detach this data sheet before commencing the examination.

This data sheet is provided for your reference.

|    |                           |                                                                                                                 |
|----|---------------------------|-----------------------------------------------------------------------------------------------------------------|
| 1  | photoelectric effect      | $E_{k\max} = hf - W$                                                                                            |
| 2  | photon energy             | $E = hf$                                                                                                        |
| 3  | photon momentum           | $p = \frac{h}{\lambda}$                                                                                         |
| 4  | de Broglie wavelength     | $\lambda = \frac{h}{p}$                                                                                         |
| 5  | resistors in series       | $R_T = R_1 + R_2$                                                                                               |
| 6  | resistors in parallel     | $\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2}$                                                                 |
| 7  | magnetic force            | $F = I l B$                                                                                                     |
| 8  | electromagnetic induction | emf : $\varepsilon = -N \frac{\Delta\Phi}{\Delta t}$ flux: $\Phi = BA$                                          |
| 9  | transformer action        | $\frac{V_1}{V_2} = \frac{N_1}{N_2}$                                                                             |
| 10 | AC voltage and current    | $V_{\text{RMS}} = \frac{1}{\sqrt{2}} V_{\text{peak}} \quad I_{\text{RMS}} = \frac{1}{\sqrt{2}} I_{\text{peak}}$ |
| 11 | voltage; power            | $V = RI \quad P = VI$                                                                                           |
| 12 | transmission losses       | $V_{\text{drop}} = I_{\text{line}} R_{\text{line}} \quad P_{\text{loss}} = I_{\text{line}}^2 R_{\text{line}}$   |
| 13 | mass of the electron      | $m_e = 9.1 \times 10^{-31} \text{ kg}$                                                                          |
| 14 | charge on the electron    | $e = -1.6 \times 10^{-19} \text{ C}$                                                                            |
| 15 | Planck's constant         | $h = 6.63 \times 10^{-34} \text{ J s}$<br>$h = 4.14 \times 10^{-15} \text{ eV s}$                               |
| 16 | speed of light            | $c = 3.0 \times 10^8 \text{ m s}^{-1}$                                                                          |

### Detailed study 3.1 – Synchrotron and applications

|    |                                                                    |                           |
|----|--------------------------------------------------------------------|---------------------------|
| 17 | energy transformations for electrons in an electron gun (<100 keV) | $\frac{1}{2} m v^2 = eV$  |
| 18 | radius of electron beam                                            | $r = p/eB$                |
| 19 | force applied to an electron beam                                  | $F = evB$                 |
| 20 | Bragg's law                                                        | $n\lambda = 2d\sin\theta$ |
| 21 | electric field between charged plates                              | $E = \frac{V}{d}$         |

**Detailed study 3.2 – Photonics**

|    |                 |                           |
|----|-----------------|---------------------------|
| 22 | band gap energy | $E = \frac{hc}{\lambda}$  |
| 23 | Snell's law     | $n_1 \sin i = n_2 \sin r$ |

**Detailed study 3.3 – Sound**

|    |                                 |                                                                                                                                             |
|----|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| 24 | speed, frequency and wavelength | $v = f\lambda$                                                                                                                              |
| 25 | intensity and levels            | sound intensity level<br>$(\text{in dB}) = 10 \log_{10} \left( \frac{I}{I_0} \right)$<br>where $I_0 = 1.0 \times 10^{-12} \text{ W m}^{-2}$ |

**Prefixes/Units**p = pico =  $10^{-12}$ n = nano =  $10^{-9}$  $\mu$  = micro =  $10^{-6}$ m = milli =  $10^{-3}$ k = kilo =  $10^3$ M = mega =  $10^6$ G = giga =  $10^9$ t = tonne =  $10^3 \text{ kg}$ **END OF DATA SHEET**